

Project Description

- This project, *Bank Loan Case Study*, focuses on understanding which factors influence whether a customer will repay or default on a bank loan.
- The goal was to study the loan application data, explore relationships between different customer and loan features, and find useful insights that can help a bank make better loan decisions.
- All the analysis was done using Microsoft Excel, mainly through data cleaning, exploratory data analysis, and visualization.

Approach

- The dataset has the file previous_application,application_data,column description.
- In this previous_application,application_data are important for analyze the data.
- I used the approach eda
 1. Data Cleaning
 2. Univariate Analysis
 3. Segmented Univariate
 4. Bivariate Analysis
 5. Outlier Detection
 6. Data Imbalance Check
 7. Correlation Analysis
 8. Visualization

Business Objectives

- It is crucial to understanding the risk factors in the bank
- We can't give a loan to who can't pay back a loan
- To analyze this I will use previous data
 - Age: Extremely young (<21) or old (>65) borrowers may have higher risk.
 - Employment Status: Permanent jobs or stable businesses are preferred.
 - Dependents: Fewer dependents = higher repayment capacity.
 - Income Level (AMT_INCOME_TOTAL): Higher, stable income → lower default risk.
 - Existing Debts: Banks check existing EMIs and credit card dues.
 - Loan-to-Income Ratio: Total loan requested ÷ income — should be reasonable (<40%).

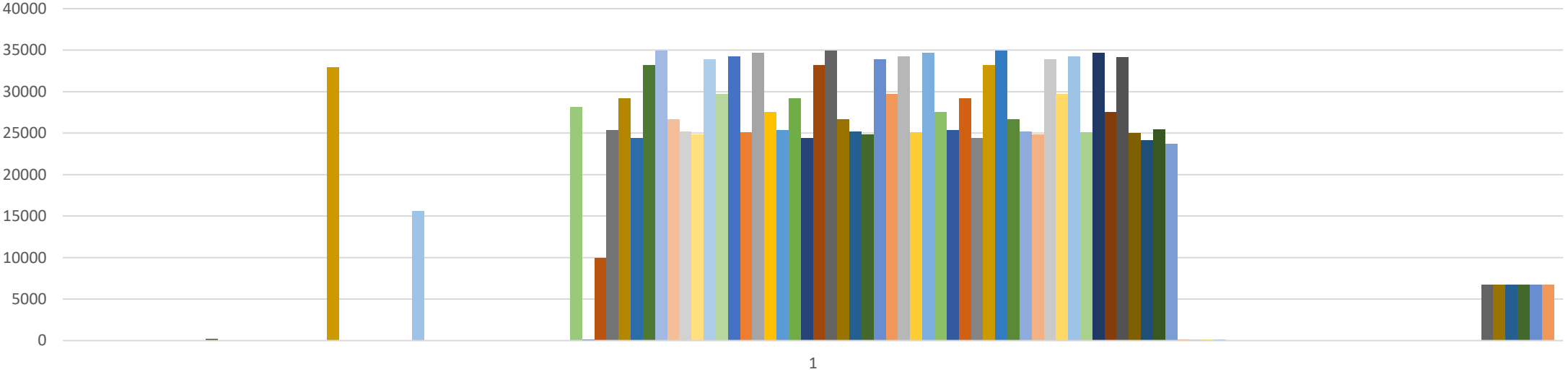
- Loan Amount (AMT_CREDIT): Should be proportionate to income and assets.
- Annuity (EMI) (AMT_ANNUIITY): Monthly payment — too high = repayment stress.
- Loan Purpose: Productive vs. consumption
- Credit Score: Based on past repayment behavior
- Past Defaults: Any missed payments or charge-offs are big red flags.
- Previous Application Records: Consistent behavior in past loans

Data Analytics Tasks:

Identify Missing Data and Deal with it Appropriately:

- Found the blanks in the application_data
- Based on the columns types I divided them into categories
columns, numeric columns, binary columns
- Numeric columns if blanks are less than 30% it is replaced by median of the column
- Categoricals are replaced by mode of columns
- Binary columns if count of 0 is more it is replaced by 0 if not 1

Missing data



- | | | | | |
|------------------------------|-----------------------------|----------------------------|-------------------------|------------------------------|
| SK_ID_CURR | TARGET | NAME_CONTRACT_TYPE | CODE_GENDER | FLAG_OWN_CAR |
| FLAG_OWN_REALTY | CNT_CHILDREN | AMT_INCOME_TOTAL | AMT_CREDIT | AMT_ANNUIITY |
| AMT_GOODS_PRICE | NAME_TYPE_SUITE | NAME_INCOME_TYPE | NAME_EDUCATION_TYPE | NAME_FAMILY_STATUS |
| NAME_HOUSING_TYPE | REGION_POPULATION_RELATIVE | DAYS_BIRTH | DAYS_EMPLOYED | DAYS_REGISTRATION |
| DAYS_ID_PUBLISH | OWN_CAR_AGE | FLAG_MOBIL | FLAG_EMP_PHONE | FLAG_WORK_PHONE |
| FLAG_CONT_MOBILE | FLAG_PHONE | FLAG_EMAIL | OCCUPATION_TYPE | CNT_FAM_MEMBERS |
| REGION_RATING_CLIENT | REGION_RATING_CLIENT_W_CITY | WEEKDAY_APPR_PROCESS_START | HOUR_APPR_PROCESS_START | REG_REGION_NOT_LIVE_REGION |
| REG_REGION_NOT_WORK_REGION | LIVE_REGION_NOT_WORK_REGION | REG_CITY_NOT_LIVE_CITY | REG_CITY_NOT_WORK_CITY | LIVE_CITY_NOT_WORK_CITY |
| ORGANIZATION_TYPE | EXT_SOURCE_1 | EXT_SOURCE_2 | EXT_SOURCE_3 | APARTMENTS_AVG |
| BASEMENTAREA_AVG | YEARS_BEGINEXPLUATATION_AVG | YEARS_BUILD_AVG | COMMONAREA_AVG | ELEVATORS_AVG |
| ENTRANCES_AVG | FLOORSMAX_AVG | FLOORSMIN_AVG | LANDAREA_AVG | LIVINGAPARTMENTS_AVG |
| LIVINGAREA_AVG | NONLIVINGAPARTMENTS_AVG | NONLIVINGAREA_AVG | APARTMENTS_MODE | BASEMENTAREA_MODE |
| YEARS_BEGINEXPLUATATION_MODE | YEARS_BUILD_MODE | COMMONAREA_MODE | ELEVATORS_MODE | ENTRANCES_MODE |
| FLOORSMAX_MODE | FLOORSMIN_MODE | LANDAREA_MODE | LIVINGAPARTMENTS_MODE | LIVINGAREA_MODE |
| NONLIVINGAPARTMENTS_MODE | NONLIVINGAREA_MODE | APARTMENTS_MEDI | BASEMENTAREA_MEDI | YEARS_BEGINEXPLUATATION_MEDI |

B. Identify Outliers in the Dataset:

- As the data is huge it is need to find outliers
- Min ,max,median,stdev,corr,q1,13,lowerbound,upper bound are found in by uing power query and had separate sheet for it for every cloum
- And vizuation is done l the exectl

C. Analyze Data Imbalance

- For this count 0, count1, proportion 0, proportion 1, and imbalance or zero are calculated by excel function for binary one
- In this flag column as more imbalance ratio
- The table in the given excel sheet .

Perform Univariate, Segmented Univariate, and Bivariate Analysis

- For the target is tasken is analyze one the sheets the different cloums are taken and anylzed based on that
- The above anaizizs are performed by using pivote table
- And created a the pie charts

Identify Top Correlations for Different Scenarios:

- The above task is done with help of excel functions and certified the hat map
- Remainig can see in the execl sheet



Tech-Stack Used

- Microsoft Excel 2023– used for data cleaning, calculations, pivot tables, and charts.
- Power Query – for filtering, removing duplicates, and cleaning missing values.
- Excel Functions: COUNT, AVERAGE, MEDIAN, CORREL, COUNTIF, IF, etc.
- VBA (Visual Basic for Applications): to automate repetitive summary calculations.
- Charts & Visualization Tools: histograms, scatter plots, and bar charts to visualize trends.

Insights

- There was a strong class imbalance — most customers had repaid their loans, while a smaller portion had defaulted.
- Income, credit amount, and annuity values varied widely, showing that customers came from very different financial backgrounds.
- Younger applicants tended to have a slightly higher chance of defaulting.
- People with lower income and more dependents were more likely to face payment difficulties.
- Missing or incomplete employment data often appeared in high-risk applications.
- The loan-to-income ratio emerged as one of the most useful indicators of default risk.

Result

- By the end of the project, I had a clear understanding of how different factors affect loan repayment behavior.
The Excel setup now provides an easy way to:
- Summarize data automatically
- Check for missing or imbalanced data
- Identify top correlations with the target variable.
- This project helped me understand how data analysis can directly support real-world banking decisions.
It also strengthened my Excel and data interpretation skills while showing how numbers translate into practical insights for credit risk assessment.

This is link to excel file

- <https://docs.google.com/spreadsheets/d/1dkr2mgsfDjncHtAAftjNx8ArbNCHvLCn/edit?usp=sharing&ouid=114401290432215454951&rtpof=true&sd=true>